Detecting Power Plant NO_x Emissions Changes using Physics-Guided Deep Learning and TEMPO Satellite Imagery

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Abstract

NASA's Tropospheric Emissions: Monitoring of Pollution (TEMPO) instrument provides hourly daytime observations of tropospheric nitrogen dioxide (NO_2) across North America. These observations could support independent, facility-level monitoring of power plant nitrogen oxides (NO_x) emissions, filling gaps caused by continuous emissions monitoring system (CEMS) failures and covering facilities not subject to hourly emissions-reporting requirements. Previous studies have estimated hourly point-source emissions from individual satellite scenes using empirical methods such as cross-sectional flux and the directional derivative approach. However, these estimates are sensitive to retrieval uncertainty, changing atmospheric transport, TEMPO's coarse spatial resolution, signals from nearby emitters, and temporal mismatch with hourly CEMS totals. We instead formulate facility monitoring as change detection over sequences of TEMPO observations. This relative target uses repeated scans to capture plume evolution while reducing sensitivity to persistent background levels and biases in absolute column values. We develop a physics-guided convolutional recurrent neural network (CRNN) that processes five consecutive TEMPO observations, High-Resolution Rapid Refresh (HRRR) meteorological fields, and transport residuals. The network jointly classifies the target power plant's emissions as increasing, steady, or decreasing and predicts a spatial map of contributions from emitting facilities within the observed scene. On a held-out evaluation set containing more than 20,000 samples from over 100 U.S. power plants observed between August 2023 and July 2026, the model achieves a classification accuracy of $X\%$, outperforming a seasonal-only baseline ($Y\%$) and a cross-sectional flux baseline ($Z\%$). These results support the use of sequential TEMPO imagery for detecting short-term changes in power plant emissions across facilities and observing conditions.

1 Introduction

Based on U.S. Environmental Protection Agency (EPA) inventory data, power plants in the U.S. electricity-generation sector emitted approximately 0.74 million short tons of NO_x in 2023, about 9.5% of national emissions [15]. NO_x emissions contribute to ground-level ozone and secondary particulate matter, pollutants that can inflame and damage the airways, reduce lung function, and aggravate asthma [13]. EPA regulations require most large U.S. fossil-fuel-fired generating units that produce electricity for sale to submit unit-level hourly emissions data on a quarterly basis [16]. However, monitoring equipment malfunctions and maintenance can make CEMS measurements unavailable or invalid, requiring plants to report calculated substitute values instead of direct measurements [14]. Behind-the-meter generators that do not sell electricity can fall outside federal hourly reporting requirements, as illustrated by the natural-gas turbines powering SpaceXAI's Colossus 2 data center [6]. Outside the United States, comparable public hourly, unit-level emissions data are generally unavailable, even where facilities continuously monitor emissions [7]. Independent estimates could therefore fill reporting gaps and reveal short-term changes before the EPA's quarterly release of hourly emissions data.

TEMPO is an ultraviolet-visible spectrometer hosted aboard a geostationary satellite that observes tropospheric vertical column densities (TVCDs) for various trace gases across North America during daylight hours [18]. Earlier polar-orbiting instruments, including the Ozone Monitoring Instrument (OMI) and TROPOspheric Monitoring Instrument (TROPOMI), typically observe a given location once per day; TEMPO's geostationary view provides approximately hourly scans with finer spatial resolution [8, 17, 18]. Hourly sampling captures intraday changes in power plant plumes and atmospheric transport that daily observations can miss. However, power plant signals in TEMPO NO_2 TVCDs can be weakened or obscured by factors including evolving meteorology and atmospheric chemistry, cloud cover, viewing geometry and retrieval uncertainty, dilution within TEMPO's 2.0-by-4.75 km spatial resolution per pixel, and interference from nearby emission sources [6, 7, 12]. Cross-comparisons with CEMS data, which is the most reliable ground truth available, introduce an additional temporal mismatch because TEMPO observes a facility at a specific time during each hourly scan, whereas CEMS data aggregate emissions over an entire clock hour [12, 16].

Satellite-based point-source studies typically estimate emissions by relating an observed column enhancement to atmospheric transport. The cross-sectional flux (CSF) method integrates background-subtracted columns across plume transects, while the integrated mass enhancement (IME) method converts the total excess plume mass to a source rate using an estimated residence time [9, 10]. Gaussian plume methods fit a parameterized dispersion model, whereas flux divergence and directional derivative approaches (DDA) apply mass-conservation relationships to satellite columns and meteorological fields [10, 11]. Santaren et al. compared five inversion methods on synthetic daily satellite images of three power plants, for which the source emissions were known, and tested the effects of cloud gaps and wind uncertainty. Gaussian plume and CSF estimates had the lowest annual-emission errors, while missing observations reduced the number of usable estimates for all methods [10]. Sun et al. applied DDA to individual hourly TEMPO NO_2 scans, integrated the resulting fields over fixed footprints around 14 U.S. power plants, and compared hourly, daily, and monthly estimates with CEMS. Adding the observed column-tendency term improved correlations at 10 plants, but the inferred rates remained

about four times lower than CEMS because of retrieval and near-source NO_x-to-NO_2 conversion uncertainties [11]. Beaudry et al. instead fit a rise–decay model to TEMPO NO_2 plumes at 64 plants and compared the resulting average emissions with CEMS; estimates fell within 50% of CEMS at 27 plants but showed substantial low bias below 250 kg NO_2 h^{-1} [2].

Machine-learning approaches offer an alternative to prescribing a fixed relationship between plume structure, atmospheric transport, and source emissions. Prior studies have addressed different targets. Finch et al. trained a convolutional neural network on 6,000 labeled TROPOMI NO_2 images to classify whether a scene contained a plume, achieving greater than 90% test accuracy without estimating its source rate [5]. Huang and Wang first labeled TROPOMI observations by whether a power plant plume was detectable, then predicted those labels from plant, meteorological, surface, and sensor features; their classifiers achieved F1 scores above 0.66 but excluded plants near major interfering sources [7]. Alnaim et al. trained recurrent models on one year of TROPOMI, meteorological, and EPA data to predict plant emissions, but focused on isolated facilities and did not evaluate generalization to previously unseen plants [1]. For other trace gases, U-Plume used a U-Net plume mask, satellite methane imagery, and wind speed to predict source rates, but its CNN estimates regressed toward the training-set mean [3]. A separate image-to-scalar CNN trained on simulated CO_2 and wind fields reached about 20% relative error when evaluated on three held-out power plants, although its evaluation did not include the retrieval errors present in real scenes [4]. To our knowledge, no published study has trained an ML model to infer facility-level emissions or behavior directly from sequences of TEMPO NO_2 imagery.

Estimating absolute hourly emissions from individual TEMPO scenes requires separating weak facility signals from background pollution and resolving uncertainties in retrieval, atmospheric chemistry, and transport. We instead use TEMPO's hourly sampling to detect relative changes over short observation sequences, reducing dependence on absolute column values and persistent background levels.

Our contributions are as follows:

- We formulate satellite-based power plant monitoring as a three-class change-detection task that identifies increasing, steady, and decreasing emissions behavior.
- We introduce a physics-guided deep learning framework that combines sequences of TEMPO NO_2 observations with meteorology and transport information. The model jointly predicts emissions behavior and a spatial map of emissions contributions within the observed scene.
- We evaluate the model on held-out power plants across geographic regions, seasons, and operating profiles and compare its performance with seasonal-only and empirical baselines.

2 Data Collection and Dataset Construction

2.1 TEMPO NO_2 Data

NASA distributes TEMPO NO_2 retrievals in two formats: Level 2 data on the instrument's native measurement footprints and Level 3 data regridded from each east-to-west scan [12]. We downloaded rasters from the version 4 level 2 product over the months covering August 2023 to August 2026. We chose the Level 2 product because it records an observation time for each mirror step, allowing us to later interpolate hourly CEMS records to the approximate time TEMPO scanned each facility.

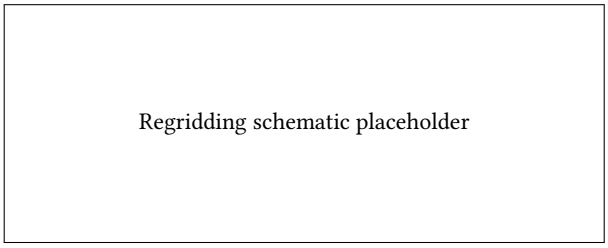


Figure 1: Planned illustration of native TEMPO Level 2 footprints mapped to the plant-centered 24×24 analysis grid.

TEMPO Level 2 observations lie on irregular native footprints, whereas our model requires a fixed spatial grid. For each power plant, we define a 72 km × 72 km area of interest (AOI) centered on the facility and divide it into a 24×24 grid of 3 km × 3 km cells. We assign each grid cell an NO_2 TVCD by averaging the valid TEMPO observations that overlap it, weighted by their area of overlap.

During regridding, only native TEMPO pixels with a main data quality flag of zero, an effective cloud fraction no greater than 0.20, and a finite NO_2 value contribute to each grid-cell average. We then create a quality mask indicating which cells in the regridded raster contain valid NO_2 values. A pretrained model later fills the masked cells.

2.2 CEMS Data

We obtained unit-level hourly emissions and operating records and annual facility and unit attributes from the U.S. Environmental Protection Agency's Clean Air Markets Program Data (CAMPD) APIs for August 2023 through September 2026.

For each AOI-hour, we summed the unit-level NO_x mass reported by all facilities in the AOI and discarded the hour if any contributing measurement was invalid or unavailable. We scored each AOI using its median total NO_x mass across hours whose mean unit operating time met or exceeded the AOI-specific median. We retained the highest-scoring 50% to focus the dataset on emissions signals more likely to be detected by TEMPO.

2.3 HRRR Weather Data

The High-Resolution Rapid Refresh (HRRR) is an hourly weather model with a 3 km grid covering the contiguous United States. From NOAA's public HRRR archive, we downloaded hourly 2 m air temperature and eastward and northward wind components at 80 m above ground for August 2023 through August 2026. We bilinearly interpolated these fields onto the same 24×24 pixel AOIs we compute from CEMS data and TEMPO observations.

2.4 Sample Construction

Each sample contains five consecutive observations of one AOI, indexed from t_0 to t_4. We retain a sequence only when each pair of

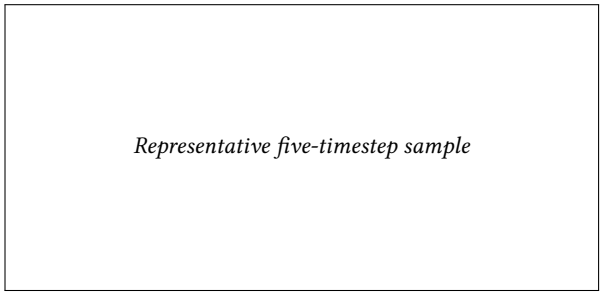


Figure 2: Representative sample with five NO_2 and meteorological observations, four transport residuals, quality masks, and spatial targets.

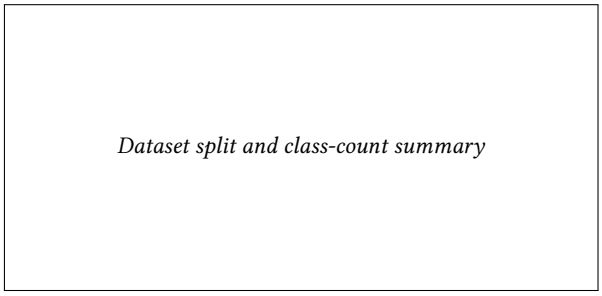


Figure 3: AOI and sample counts by split, with class composition before and after sampling.

adjacent TEMPO scans is separated by 40–70 minutes and every NO_2 raster has valid values in at least 85% of its cells. At each timestep, we pair the TEMPO raster with the HRRR temperature and wind rasters from the closest hour.

Because CEMS records correspond to clock hours rather than TEMPO scan times, we linearly interpolate the AOI-hour NO_x emissions between the surrounding hourly CEMS records at each scan time. Let E_i denote the interpolated emissions at time t_i. For scans t_a through t_b, we calculate the time-weighted mean

$$\overline{E}_{a:b} = \frac{\sum_{i=a}^{b-1}(t_{i+1} - t_i)(E_i + E_{i+1})/2}{t_b - t_a}. \tag{1}$$

We define the sequence change as $\Delta E = \overline{E}_{2:4} - \overline{E}_{0:2}$. A sequence is labeled as an increase when $\Delta E \geq 250\ \mathrm{lb\,h^{-1}}$, a decrease when $\Delta E \leq -250\ \mathrm{lb\,h^{-1}}$, and steady otherwise.

To produce transport residuals for each pair of adjacent NO_2 rasters, we use a semi-Lagrangian transport model to advect the earlier raster with the average of the two endpoint wind fields while accounting for horizontal dispersion and effective NO_2 decay. We subtract the transported estimate from the next TEMPO observation to isolate new emission signal from the later timestep not captured by the existing plume. We retain the original NO_2 quality masks and combine them after transport to identify unreliable residual pixels.

For each sample, we also compute increase and decrease pixel contribution rasters (sized 24×24) to allow the model to learn plume attribution. We first sum unit emissions within each facility, interpolate each facility's emissions at the five scan times, and calculate its time-weighted change using the same early and late windows from above. For an increase sample, positive facility changes are normalized to sum to one and distributed around their facility locations on the increase map. Decrease samples use the normalized magnitudes of negative facility changes on the decrease map. The inactive map is zero, and both maps are zero for steady samples.

2.5 Dataset Stratification

To prevent dataset leakage, we group overlapping AOIs into clusters and strictly assign every AOI in a common cluster to the same dataset split. A mixed-integer optimization assigns clusters to the training, validation, and test sets by minimizing deviations from 70%, 15%, and 15% split targets for both AOI and sample counts. After fixing the training clusters, we divide the remaining clusters between validation and test while also balancing their class counts.

Steady class samples are more common than emission changes, and increase samples are more common than decrease samples. We therefore downsample each training class to the size of the smallest class. For validation and test, we retain every increase and decrease sequence and sample steady sequences to match the larger change class. This preserves the imbalance between increases and decreases in the held-out sets while preventing steady behavior from dominating evaluation.

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